

Practice Questions for the Students for the Examination

Bachelor Level / Third Year / Fifth Semester

Computer Science and Information Technology (CSC316)

Full Marks: 60

Pass Marks: 24

Time: 3 Hours

Model Question Set 1

Section A

Attempt any TWO questions.

(2×10 = 20)

1. (a) Explain Feistel cipher structure with neat diagram. [4]
(b) Describe DES key generation and encryption steps (high level). [4]
(c) What is Meet-in-the-Middle attack in Double DES? Why Triple DES is used? [2]
2. (a) List any four properties and two applications of cryptographic hash functions. [3]
(b) Explain how SHA-1 generates a 160-bit hash (message padding, parsing, main loop overview). [7]
3. In RSA, a user chooses $p = 11$, $q = 17$.
(a) Compute n , $\phi(n)$ and select $e = 7$. Find d . [5]
(b) Using mapping $a = 0, \dots, z = 25$, encrypt the plaintext “hi” and show decryption. [5]

Section B

Attempt any EIGHT questions.

(8×5 = 40)

4. Differentiate between security policy and security mechanism with an example. [5]
5. Decrypt the ciphertext “KQNV” using Hill cipher with key $\begin{bmatrix} 3 & 5 \\ 7 & 2 \end{bmatrix} \pmod{26}$. Show steps. [5]
6. Find the multiplicative inverse of 17 mod 43 using Extended Euclidean Algorithm. [5]

7. Explain AES key expansion concept and why round keys are needed. [5]
8. Distinguish ECB vs CBC modes and mention one weakness of ECB. [5]
9. Define digital certificate and write stages of certificate life cycle. [5]
10. List services provided by PGP for secure email. [5]
11. Differentiate virus, worm, trojan horse, zombie (any three with clear points). [5]
12. Write short notes on any TWO: [2.5+2.5]
 - (a) Quadratic residue
 - (b) Discrete logarithm
 - (c) Stateful inspection firewall

Model Question Set 2

Section A

Attempt any TWO questions.

(2×10 = 20)

1. (a) Explain IDEA algorithm round structure (operations and subkeys). [6]
(b) Why is IDEA considered strong? Mention design ideas (mixing operations). [4]
2. Over $GF(7)$:
 - (a) Divide $6x^2 + 5x + 4$ by $3x + 2$. [6]
 - (b) Define $GF(p)$ and explain why arithmetic is done modulo p . [4]
3. In ElGamal, let $p = 101$, $g = 2$, and receiver public key $y = 35$.
 - (a) Encrypt message $m = 45$ using $k = 9$ (show formula and computed pair). [6]
 - (b) Explain decryption steps and why randomness k is necessary. [4]

Section B

Attempt any EIGHT questions.

(8×5 = 40)

4. Encrypt “security” using Vigenere cipher with key “CODE”. Show method briefly. [5]
5. Construct Playfair matrix with keyword “DESTINY” and encrypt “NETWORK”. [5]
6. Explain Miller–Rabin primality test and test whether $n = 91$ is prime/composite (show one witness). [5]
7. Define Euler Totient function $\phi(n)$ and compute $\phi(45)$. [5]
8. Explain Diffie–Hellman key exchange with a small example (choose p , g , private keys). [5]
9. What is Man-in-the-Middle attack in DH? Draw message flow and explain. [5]
10. Explain Needham–Schroeder scheme (goal and main steps). [5]

11. List any four types of firewall and one use-case each. [5]
12. Short notes on any TWO: [2.5+2.5]
- (a) One-time pad
 - (b) X.509 certificate fields
 - (c) Rule-based intrusion detection

Model Question Set 3

Section A

Attempt any TWO questions. $(2 \times 10 = 20)$

1. (a) Explain substitution vs transposition ciphers with examples. [4]
(b) Encrypt “COMPUTER” using Rail Fence cipher with 3 rails. [3]
(c) Why are classical ciphers insecure against modern attacks? [3]
2. (a) Explain AES encryption process (SubBytes, ShiftRows, MixColumns, AddRoundKey). [6]
(b) Demonstrate one MixColumns computation for one column (use any example bytes). [4]
3. (a) Define digital signatures and explain Direct vs Arbitrated digital signature. [4]
(b) Explain RSA-based signature generation and verification with steps. [6]

Section B

Attempt any EIGHT questions. $(8 \times 5 = 40)$

4. Define CIA triad and explain with one real example in network security. [5]
5. Decrypt “UQ” using Caesar cipher with shift 7 (show steps). [5]
6. Encrypt “data” using Vernam cipher with key “code” (show XOR / modular add method). [5]
7. In Z_{26} , find additive inverse of 19 and multiplicative inverse of 15 (if exists). [5]
8. Define primitive root and check whether 2 is primitive root of 11. [5]
9. Explain message authentication code (MAC) and its purpose. [5]
10. Compare MD5 vs SHA-1 (output size, structure, security status). [5]
11. Explain challenge–response authentication with a message flow diagram. [5]
12. Short notes on any TWO: [2.5+2.5]
 - (a) Dictionary attack (online vs offline)
 - (b) Non-repudiation
 - (c) Clandestine intruder

Model Question Set 4

Section A

Attempt any TWO questions. $(2 \times 10 = 20)$

1. (a) Explain DES encryption steps and role of S-Box. [5]
(b) What are weak keys in DES? Why are they problematic? [3]
(c) Compare Double DES vs Triple DES. [2]
2. (a) Explain modes of block cipher: ECB, CBC, CFB, OFB, CTR (any four). [6]
(b) For CBC mode, show formula for encryption and decryption and explain role of IV. [4]
3. (a) Explain IPSec architecture and difference between AH and ESP. [6]
(b) Explain tunnel mode vs transport mode with use cases. [4]

Section B

Attempt any EIGHT questions. $(8 \times 5 = 40)$

4. Differentiate between stream cipher and block cipher with examples. [5]
5. Perform $3^{202} \bmod 11$ using Fermat/Euler theorem (show steps). [5]
6. Explain discrete logarithm problem and why it matters in cryptography. [5]
7. In RSA, explain why padding is important (conceptual answer). [5]
8. Explain SSL/TLS handshake at a high level (who sends what). [5]
9. Write key points of SSL Record Protocol security (confidentiality and integrity). [5]
10. Explain PKI trust models (hierarchical vs web-of-trust). [5]
11. Define intrusion and explain statistical anomaly detection IDS. [5]
12. Short notes on any TWO: [2.5+2.5]
 - (a) Proxy firewall
 - (b) Masquerader intruder
 - (c) Nonce in authentication

Model Question Set 5

Section A

Attempt any TWO questions. $(2 \times 10 = 20)$

1. (a) Explain message authentication and need of MAC. [4]
(b) Explain MD4/MD5 message digest process (padding, blocks, rounds overview). [6]
2. (a) Explain DSS approach for digital signature (DSA idea and parameters). [6]
(b) Compare DSA vs RSA signatures (any four points). [4]
3. (a) Explain Kerberos protocol and role of TGT. [6]
(b) Explain how Kerberos helps in mutual authentication. [4]

Section B

Attempt any EIGHT questions. $(8 \times 5 = 40)$

4. Define authentication system and classify authentication factors (knowledge/possession/inherence) [5]
5. Explain biometric system: working steps and two issues (FAR/FRR). [5]
6. A user claims $Z_{10}^* = \{1, 3, 7, 9\}$. Explain why these are the units and verify with gcd. [5]
7. Find GCD(252, 198) using Euclidean algorithm and interpret in cryptography context. [5]
8. Explain polynomial multiplication over $GF(2)$ with a small example. [5]
9. Define X.509 certificate and list any six important fields. [5]
10. Explain next-generation firewall features vs traditional packet filtering. [5]
11. Explain DoS attack and give two mitigation ideas. [5]
12. Short notes on any TWO: [2.5+2.5]
 - (a) Replay attack
 - (b) Trojan horse
 - (c) Counter (CTR) mode

Model Question Set 6

Section A

Attempt any TWO questions.

(2×10 = 20)

1. RSA (complete numerical). Let $p = 23$, $q = 29$.
 - (a) Compute n , $\phi(n)$ and choose $e = 5$. Find private key d . [5]
 - (b) Using mapping $a = 0, \dots, z = 25$, encrypt plaintext “cs” and then decrypt to verify. [5]
2. Finite field / modular arithmetic.
 - (a) Using Extended Euclidean Algorithm, find $37^{-1} \pmod{101}$. [4]
 - (b) Find the set of quadratic residues modulo 17. [3]
 - (c) Over $GF(7)$, divide $6x^3 + 2x^2 + 5x + 1$ by $3x + 4$ (quotient and remainder). [3]
3. Diffie–Hellman + attack. Consider $p = 47$, $g = 5$. Alice chooses $a = 9$, Bob chooses $b = 13$.
 - (a) Compute $A = g^a \pmod{p}$ and $B = g^b \pmod{p}$. [4]
 - (b) Compute the shared secret key K . [3]
 - (c) Show how Man-in-the-Middle attack can break DH (message flow and keys created). [3]

Section B

Attempt any EIGHT questions.

(8×5 = 40)

4. Decrypt “KQNV” using Hill cipher with key $\begin{bmatrix} 11 & 8 \\ 3 & 7 \end{bmatrix} \pmod{26}$. Show steps. [5]
5. Use Miller–Rabin with base $a = 2$ to show whether 341 is prime or composite. [5]
6. Compute $7^{222} \pmod{13}$ using Euler/Fermat theorem (show steps). [5]
7. Find $\phi(84)$ and then compute $5^{\phi(84)} \pmod{84}$. [5]
8. Verify whether 3 is a primitive root modulo 17. [5]

9. ElGamal (encryption only): $p = 101$, $g = 2$, $y = 38$, $k = 7$, $m = 45$. Compute (c_1, c_2) . [5]
10. AES MixColumns (partial numerical). Given column $[a_0, a_1, a_2, a_3] = [57, 83, 1A, C6]$ (hex), compute $b_0 = 02 \cdot a_0 \oplus 03 \cdot a_1 \oplus a_2 \oplus a_3$ using xtime rule. [5]
11. CBC mode with toy cipher $E_k(x) = x \oplus k$. Let $k = 3A$, $IV = F0$, plaintext blocks: 12, 34, 56. Compute CBC ciphertext blocks C_1, C_2, C_3 . [5]
12. Over $GF(2)$, multiply $(x^3 + x + 1)(x^2 + x + 1)$. [5]

Model Question Set 7

Section A

Attempt any TWO questions. $(2 \times 10 = 20)$

1. ElGamal (encryption + decryption). Let $p = 103$, $g = 5$. Receiver private key $x = 19$.
 - (a) Compute public key $y = g^x \bmod p$. [3]
 - (b) Sender encrypts message $m = 72$ using $k = 11$. Compute (c_1, c_2) . [4]
 - (c) Show decryption to recover m . [3]
2. Toy IDEA Round Trace (4-bit words). Work in 4-bit words (0–15). Define: Addition = mod 16, XOR = bitwise, Multiplication = mod 17 with 0 treated as 16. Input block $(X_1, X_2, X_3, X_4) = (9, 4, 12, 3)$. Subkeys: $Z_1 = 7$, $Z_2 = 5$, $Z_3 = 11$, $Z_4 = 9$, $Z_5 = 6$, $Z_6 = 10$. Trace Round-1 completely and write output (Y_1, Y_2, Y_3, Y_4) . [10]
3. CBC-MAC using toy cipher $E_k(x) = x \oplus k$. Let $k = 9C$, $IV = 00$. Message blocks: $6D, A1, 3F, 20$.
 - (a) Compute CBC-MAC tag. [6]
 - (b) Explain why CBC-MAC is insecure for variable-length messages (one attack idea). [4]

Section B

Attempt any EIGHT questions. $(8 \times 5 = 40)$

4. Using EEA, compute $91^{-1} \bmod 143$ (if exists). If not, justify using gcd. [5]
5. Over $GF(7)$, compute $(4x^2 + 6x + 1) + (5x^2 + 2x + 6)$ and $(4x^2 + 6x + 1)(5x + 3)$. [5]
6. Compute discrete log: find x such that $2^x \equiv 18 \pmod{29}$. [5]
7. Use Fermat's theorem to compute $3^{1000} \bmod 31$. [5]
8. Encrypt "cryptography" using key "NEPAL" (show first 6 steps clearly). [5]

9. OTP/Vernam (XOR): plaintext byte B6, key byte 3D. Compute ciphertext and decrypt back. [5]
10. RSA signature (toy): $p = 17$, $q = 19$, choose $e = 5$. Compute d , sign $m = 10$, and verify ($s = m^d \bmod n$, $m' = s^e \bmod n$). [5]
11. SHA-1 padding: for message length 1000 bits, compute (a) number of padding bits, (b) total length after padding, (c) number of 512-bit blocks. [5]
12. Explain and compare ECB vs CTR mode, and state why CTR is parallelizable. [5]

Model Question Set 8

Section A

Attempt any TWO questions.

(2×10 = 20)

1. Hill Cipher 3×3 (advanced numerical). Key matrix (mod 26):

$$K = \begin{bmatrix} 2 & 4 & 5 \\ 9 & 2 & 1 \\ 3 & 17 & 7 \end{bmatrix}$$

- (a) Show that K is invertible in Z_{26} (det and gcd check). [3]
 - (b) Encrypt plaintext “attacknow” (pad if needed). [4]
 - (c) Compute K^{-1} (mod 26) and decrypt your ciphertext to verify. [3]
2. AES $GF(2^8)$ numerical (xtime + MixColumns). Use AES rules: $02 \cdot x = \text{xtime}(x)$, $03 \cdot x = (02 \cdot x) \oplus x$. Given column [AE, 47, D1, 2C] (hex).
- (a) Compute $02 \cdot \text{AE}$ and $03 \cdot 47$ using xtime and reduction by 1B when needed. [4]
 - (b) Compute $b_0 = 02a_0 \oplus 03a_1 \oplus a_2 \oplus a_3$. [3]
 - (c) Compute $b_1 = a_0 \oplus 02a_1 \oplus 03a_2 \oplus a_3$. [3]
3. Number theory + RSA (high numerical).
- (a) Use Miller–Rabin (base $a = 2$) to test whether 221 is prime or composite. [4]
 - (b) Compute $\phi(187)$ and then compute $11^{\phi(187)} \bmod 187$. [3]
 - (c) Using RSA with $p = 11$, $q = 17$, $e = 7$, encrypt and decrypt message $m = 88$. [3]

Section B

Attempt any EIGHT questions.

(8×5 = 40)

- 4. Find multiplicative inverse of 75 mod 121 using EEA. [5]
- 5. Over $GF(7)$, divide $5x^4 + 3x^2 + 6x + 2$ by $2x^2 + 4$. [5]
- 6. Compute all units in Z_{18} and justify using gcd property. [5]

7. In Diffie–Hellman with $p = 59$, $g = 2$, if $a = 11$, $b = 17$, compute shared key. [5]
8. ElGamal: $p = 107$, $g = 2$, $x = 31$, $k = 19$, $m = 50$. Compute y, c_1, c_2 . [5]
9. Compute $9^{123} \bmod 37$ using fast exponentiation (show steps). [5]
10. Playfair: create matrix with keyword “GALOIS” and encrypt “SECURITY” (show digraph rules). [5]
11. CBC with toy cipher $E_k(x) = x \oplus k$: $k = 5F$, $IV = AA$, plaintext blocks 01, 10, 11. Compute ciphertext blocks. [5]
12. Find quadratic residues mod 11 and use them to decide if 7 is a quadratic residue mod 11. [5]

Model Question Set 9

Section A

Attempt any TWO questions. $(2 \times 10 = 20)$

1. Define CIA triad. State the encryption process of double and triple DES. What is the task of S-Box in DES? Discuss with an example. [3+2+5]
2. Explain the generic model of digital signature process. Consider the two prime numbers 7 and 19. Select 29 as public key and 41 as private key. Encrypt the plaintext 4 and decrypt the cipher text 3 using RSA. [5+5]
3. Define Galois field with an example. Explain any two modes of block cipher encryptions. Determine the quadratic residues of 7. [2+4+4]

Section B

Attempt any EIGHT questions. $(8 \times 5 = 40)$

4. What does intrusion mean? How the system detect intrusion? List any four types of firewall. [1+2+2]
5. Decrypt the message "GVPJ" using Hill cipher taking the key as $\begin{bmatrix} 3 & 7 \\ 5 & 12 \end{bmatrix}$. [5]
6. Describe the PKI trust model. [5]
7. Define an authentication system. Illustrate the need of mutual authentication over one way authentication with an example. [2+3]
8. Find the value of $7^{2019} \bmod 13$ using Fermat's Little theorem. Define Euler totient function with an example. [2+3]
9. List the properties of hash function. Discuss the first pass of MD4. [2+3]
10. Differentiate between Symmetric and Asymmetric cipher. Encrypt the message "HELL" using the key "FAIL" using Vernam cipher. [3+2]
11. Divide $3x^2 + 4x + 3$ by $5x + 6$ over $GF(7)$. [5]
12. Define SSL protocol. Mention the services provided by PGP. [2+3]

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Model Question Set 10

Section A

Attempt any TWO questions.

(2×10 = 20)

1. Illustrate the concept of security policy and mechanism with an example. Differentiate between block cipher and stream cipher. Explain the process of key expansion in AES. [3+2+5]
2. Describe the properties of hash functions. Discuss how hash value is generated using SHA-1 algorithm. [4+6]
3. Show that Z_5 is a field. John publishes the ElGamal public key $(q, \alpha, Y_A) = (101, 2, 14)$. Jane desired to send the secret message “CSIT” to John. Using the equivalence $A = 0, B = 1, \dots, Z = 25$, encrypt the message using John’s public key. Use a random number $k = 4$. [5+5]

Section B

Attempt any EIGHT questions.

(8×5 = 40)

4. Differentiate between Trojan horse and virus. Describe any two types of intruders. [2+3]
5. The message “IMOGUN” was encrypted with a Playfair cipher using keyword “GA-LOIS”. Decrypt the message. [5]
6. How encryption is done using IDEA algorithm. [5]
7. Describe the services provided by Pretty Good Privacy protocol to secure email. [5]
8. What is the condition for two integers x and y to be relatively prime? Find whether 61 is prime or not using Miller–Rabin algorithm. [1+4]
9. Define challenge response system. Why do we need Kerberos? [2+3]

10. How direct digital signature is different from arbitrated digital signature? How digital signature generation and verification is done using RSA. [2+3]
11. Which one is more secure, monoalphabetic cipher or poly alphabetic cipher? Justify. Using Rail fence cipher, encrypt the text “LEARNING AND TEACHING ARE DIFFERENT”, using 3 rails. [1+4]
12. Why do we need discrete logarithm over normal logarithm? Find out whether 3 is primitive root of 7 or not. [2+3]

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Model Question Set 11

Section A

Attempt any TWO questions.

(2×10 = 20)

1. Among monoalphabetic and polyalphabetic cipher, which one is more vulnerable? Justify your statement. Which types of keys are considered as weak keys in DES? Explain the round operation in IDEA. [2+2+6]
2. State Fermat's theorem with example. Given the prime number $p = 29$ and its primitive root $g = 8$, private key of sender with $X = 9$ and random integer $K = 11$, encrypt the message $m = 13$ using ElGamal cryptosystem. [5+5]
3. Compare the SHA parameters between SHA-1 and SHA-2 family. Decrypt the cipher text "DRI" with key $\begin{bmatrix} 7 & 8 \\ 11 & 11 \end{bmatrix}$ using Hill Cipher. [3+7]

Section B

Attempt any EIGHT questions.

(8×5 = 40)

4. Define discrete logarithm. Explain the procedure of sharing the secret key in Diffie-Hellman. [2+3]
5. Distinguish between stream cipher and block cipher. Encrypt the message "WE ARE IN SAME RACE UNTILL OJR LIVE END" using Rail fence cipher using 4 as number of rails. [2+3]
6. Define digital signature. Describe the approaches of DSS. [2+3]
7. What is the task of firewall? List the elements of X.509. [2+3]
8. How does the nature of worms differ with virus? Define PKI with its architecture model. [1+4]
9. Explain the procedure of mix column transformation in AES with an example. [5]
10. What is the role of prime number in Euler totient function? Find the GCD of 12 and 16 using Euclidean algorithm. [2.5+2.5]

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Model Question Set 12

Section A

Attempt any TWO questions.

(2×10 = 20)

1. Define discrete logarithm. How key generation, encryption and decryption is done in RSA. In a RSA cryptosystem, given $p = 13$ and $q = 7$, determine private key, public key, and perform encryption and decryption for the text $M = \text{"hi"}$ using 0 to 25 for letters from a to z. [10]
2. Write down the encryption and decryption process at 2-DES and 3-DES. Explain the Feistel cipher structure. Divide $5x^2 + 4x + 6$ by $2x + 1$ over $GF(7)$. [3+3+4]
3. What are the applications of hash functions? Discuss how SHA-1 algorithm generates hash value from a given message. [2+8]

Section B

Attempt any EIGHT questions.

(8×5 = 40)

4. Show encryption and decryption of "csit" using hill cipher having key $K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$. [5]
5. Give the formal definition of Authentication system. Describe about one way and mutual Authentication system. [2+3]
6. List the stage of certificate life cycle. What are the types of firewalls? [3+2]
7. What is malicious logic? How Zombies are different from Trojan horses? [1+4]
8. How Miller Rabin test is used for primality testing? Show whether the number 561 passes the test. [2.5+2.5]
9. Show that the set of integers is Ring under addition and multiplication. [5]

10. How substitution ciphers are different from transposition ciphers? Given message $M = \text{"CSIT PROGRAM IS A HOT CAKE"}$, encrypt M using Rail Fence cipher with rail size=3. [2+3]
11. Describe about IPSec. List the five services of PGP. [3+2]
12. How does meet in middle attack work in Diffie Hellman key Exchange protocol? Explain. [5]

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Model Question Set 13

Section A

Attempt any TWO questions.

(2×10 = 20)

1. Let us consider the 4 bits key input text as {1100, 1010, 0000, 1111, 0101, 1001} and input text as {1011, 1110, 1011, 1000}. Now trace the first full round operation of IDEA algorithm. [10]
2. What is message authentication code? List the operations of computing digest value in different passes of MD4. Describe about Needham–Schroeder protocol. [1+5+4]
3. Why do we need discrete logarithms? Illustrate with an example. Consider a Diffie–Hellman scheme with a common prime $p = 13$ between user A and B. Suppose public key of A is 10 and public key of B is 8. Now determine their private keys and shared secret key. Select any valid primitive root of 13. [3+7]

Section B

Attempt any EIGHT questions.

(8×5 = 40)

4. Show encryption of plaintext “ALGORITHM” using the key “PSEUDOCODE” using playfair cipher. [2+3]
5. Discuss the working mechanism of the Kerberos protocol. [5]
6. What is the use of firewall? How circuit level gateway differs from stateful inspection firewall? [2+3]
7. What is intrusion? Explain any two types of intrusion detection system. [1+4]
8. Find the multiplicative inverse of polynomial (95) using Extended Euclidean Algorithm. [5]
9. What is DoS attack? Discuss about PKI trust model. [2+3]

10. Using Vigenere cipher with key “worlds”, encrypt the plain text “hello everyone”.
[5]
11. Describe the different modes of block cipher encryption. [5]
12. Write short notes on any TWO. [2.5+2.5]
 - (a) Totient value of a positive integer
 - (b) Properties of hash function
 - (c) Virus vs Worms